AER1301: MOMENT CLOSURE METHODS FOR KINETIC EQUATIONS OF COMPLEX TRANSPORT PHENOMENA

Assignment #1

1. A hypersonic wind tunnel is contructed so as to obtain a high-speed flow with $M = 7$ in the test section. At that Mach number, the test section gas velocity, density, temperature, pressure, and viscosity are $V = 1743$ m/s, $\rho = 0.0182$ kg/m^3, $T = 154$ K, $p = 8.33$ kPa, and $\mu = 12.2(10)^{-6}$ kg/m-s, respectively. A model 2.54 cm high and 15.24 cm long is to be tested in the wind tunnel. A pitot tube having an outside diameter of 0.0762 mm will be used to measure the free-stream stagnation pressure in front of the model. Determine the Reynolds and Knudsen numbers for the flows about the model and probe and determine whether or not non-equilibrium rarefied flow effects should be important for the model and the probe, respectively. Assume the gas particles behave as hard spheres during collisions such that the mean free path, λ, is given by the expression

$$\lambda = \frac{16\mu}{5\rho}\frac{1}{\sqrt{2\pi RT}}\,,$$

where R is the ideal gas constant and $p = \rho RT$.

2. In a vacuum, the vector form of Maxwell's equations is the following:

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}\,,$$

$$\vec{\nabla} \times \vec{B} = \epsilon_\circ \mu_\circ \frac{\partial \vec{E}}{\partial t} + \mu_\circ \vec{j}\,,$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_\circ}\,,$$

$$\vec{\nabla} \cdot \vec{B} = 0\,,$$

where $\vec{E}$ and $\vec{B}$ are the electric and magnetic field vectors, ρ is the electric charge density, $\vec{j}$ is the electric current density, and $\mu_\circ$ and $\epsilon_\circ$ are the magnetic permeability and electric permittivity of free space, respectively. Re-express these equations using tensor notation.

3. For a gas in thermal equilibrium, find the ratios between the characteristic speeds of molecules: $< vf >:< v^2 f >^{1/2}$: v_m, where v is the molecule speed, v_m is the most probable speed of the molecules, and f is the probability density function (PDF) for the gas particle speeds.

4. Consider a stationary gas consisting of molecules of mass m in thermal equilibrium with temperature T.

   (a) Derive an expression for the probability density function (PDF) of translational kinetic energy, $f_E(E)$, where $E = mv^2/2$ and $f_E(E)dE$ is the fraction of molecules whose kinetic energy lies in the range between E and $E + dE$.

   (b) Show that $\bar{E} =< Ef_E >= 3kT/2$ and $E_m = kT/2$ where E_m is the most probable translational kinetic energy.

5. A gas has the following normalized velocity distribution function or so-called probability density function (PDF), $f = \mathcal{F}/n$:

$$f(\vec{v}) = \left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2} \left[1 + A v_x v_y + B v_z \left(v^2 - \frac{5}{2\beta}\right)\right],$$

where A and B are constants independent of the particle velocity, $\vec{v}$, $\beta = m/2kT$, T is the gas temperature, and $v^2 = |\vec{v}|^2 = v_x^2 + v_y^2 + v_z^2$. Determine $m < \vec{v} f >$, $m < (v_x^2/2) f >$, $m < (v_y^2/2) f >$, $m < (v_z^2/2) f >$, and $m < \vec{v}(v^2/2) f >$.

6. The number density function (NDF) of molecular velocities in a non-equilibrium spatially uniform monatomic gas of molecular mass, m, is given by

$$\mathcal{F}(\vec{v}) = C g_x(v_x) g_y(v_y) g_z(v_z),$$

where g_x, g_y, and g_z are triangular-shaped functions given by

$$g_x(v_x) = \begin{cases} 0 & \text{for } v_x \le -a_x \\ a_x + v_x & \text{for } -a_x < v_x \le 0 \\ a_x - v_x & \text{for } 0 < v_x < a_x \\ 0 & \text{for } v_x \ge a_x \end{cases},$$

$$g_y(v_y) = \begin{cases} 0 & \text{for } v_y \le -a_y \\ a_y + v_y & \text{for } -a_y < v_y \le 0 \\ a_y - v_y & \text{for } 0 < v_y < a_y \\ 0 & \text{for } v_y \ge a_y \end{cases},$$

$$g_z(v_z) = \begin{cases} 0 & \text{for } v_z \le -a_z \\ a_z + v_z & \text{for } -a_z < v_z \le 0 \\ a_z - v_z & \text{for } 0 < v_z < a_z \\ 0 & \text{for } v_z \ge a_z \end{cases},$$

and where C is a normalization factor and a_x, a_y, and a_z are constant, positive, real-valued quantities.

   (a) Given the number density for the gas is n, determine the value of the normalization factor C.

   (b) What is the average (bulk) velocity of the gaseous particles? Provide expressions for each component of the average velocity vector.

   (c) What is the gas pressure, p?

   (d) What is the heat flux for the gas, $\vec{q}$? Provide expressions for each component of the heat flux vector.

   Please show the steps taken to calculate the preceding values when providing your answers.

7. Show that the Boltzmann equation is invariant for all Galilean transformations. A Galillean transformation relates the position, $\vec{x}$, in a stationary inertial frame to the position in a translating reference frame, $\vec{x}'$, according to the expression $\vec{x} = \vec{x}' + \vec{u}_\circ t$ where $\vec{u}_\circ$ is the translation velocity of the moving frame relative to the stationary frame ($\vec{u}_\circ$ is taken to be a constant).

8. (a) Show that the two formulations of the conserved form of the Boltzmann equation given below are equivalent:

$$\frac{\partial \mathcal{F}}{\partial t} + v_i \frac{\partial \mathcal{F}}{\partial x_i} + a_i \frac{\partial \mathcal{F}}{\partial v_i} = \frac{\delta \mathcal{F}}{\delta t} \,,$$

and

$$\frac{\partial \mathcal{F}}{\partial t} + \vec{\nabla} \cdot (\mathcal{F}\vec{v}) + \vec{\nabla}_v \cdot (\mathcal{F}\vec{a}) = \frac{\delta \mathcal{F}}{\delta t} \,,$$

where $\mathcal{F}$ is the particle number density function (NDF), $\vec{v}$ is the velocity, $\vec{a}$ is the acceleration of the particles due to an applied external force, and $\vec{\nabla}_v \cdot \vec{A} = \partial A_i / \partial v_i$, where $\vec{A}$ is a vector quantity. Assume that the external force field, $\vec{a}$, is divergence free such that $\vec{\nabla}_v \cdot \vec{a} = 0$.

(b) Starting with the conserved form of the Boltzmann equation

$$\frac{\partial \mathcal{F}}{\partial t} + v_i \frac{\partial \mathcal{F}}{\partial x_i} + a_i \frac{\partial \mathcal{F}}{\partial v_i} = \frac{\delta \mathcal{F}}{\delta t} \,,$$

multiply this equation by $mc_j = m(v_j - u_j)$ and integrate the result over all velocity space to obtain the usual form of the momentum equation. Show all relevant steps in detail. Again assume that the external force field is divergence free.